Geography Ch-8 (Must-Do) — Limestone & Chalk (Karst) Landforms

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain
Karst by carbonation/solution: swallow holes, dolines, uvala, caverns, stalactites
CA anchor: limestone mining vs karst aquifers (Meghalaya)

Facts from official sources | Inferences clearly marked | No fabricated data

■ HIGH UPSC RELEVANCE

HIGH

FOUNDATION — Karst Erosional & Depositional Landforms (GC Leong / Majid)

GS-I · Physical Geography → Karst Landforms (Source: GC Leong, Majid Husain)

■ NEWS TRIGGER

Karst geomorphology explains India's spectacular limestone caves (Meghalaya) — and why their aquifers are so vulnerable to mining and contamination.

■ INTRO & ORIGIN

In limestone (chalk) country, the master process is **carbonation/solution**: rain + CO_2 forms weak carbonic acid that dissolves calcium carbonate. Karst thus has little surface water and no integrated drainage.

*Water sinks at **swallow/sink holes** along joints, etching out **caverns** underground; dissolved calcium carbonate is later re-deposited drip by drip as **stalactites and stalagmites**.*

■ KEY DATA

Type	Landform	What it is
Surface erosional	Clints & grikes (lapies/karren)	Blocks & solution-widened joints on bare limestone pavement
Surface erosional	Swallow/sink hole; doline	Point where water sinks; small solution/collapse hollow
Surface erosional	Uvala / Polje	Coalesced dolines / large flat-floored basin
Sub-surface erosional	Cavern, underground passage	Solution along joints & bedding planes
Depositional	Stalactite / Stalagmite / Pillar	Ceiling-down / floor-up dripstone; meeting = pillar

■ STATIC THEORY

• **Carbonation/solution** of calcium carbonate is the dominant process; karst is dry at the surface and lacks a well-integrated drainage system (Majid).

• **Swallow/sink holes** carry rain into the rock (e.g. Gaping Ghyll, Yorkshire); enlarged they merge into **dolines**, then **uvalas** (coalesced dolines) and huge **poljes**.

• **Clints & grikes** (limestone pavement): blocks (clints) separated by solution-widened joints (grikes); French term for karren is **lapies**.

• Underground, water etches **caverns** and passages along joints/bedding planes.

• **Depositional dripstone: stalactites** hang from the roof, **stalagmites** rise from the floor; when they meet they form a **pillar/column** (Batu, Mammoth, Carlsbad, Postojna caves).

■ MUST-KNOW FACTS

1. Master process = carbonation/solution ($CaCO_3$ + carbonic acid).

2. Karst = little surface water, disintegrated drainage.

3. Sink/swallow hole -> doline -> uvala -> polje (size order).

■■ UPSC TRAPS

✗ Stalagmites hang from the cave roof.

✓ Stalactites hang from the ROOF; stalagmites rise from the FLOOR.

✗ Karst regions have dense surface drainage.

4. Clints (blocks) + grikes (solution-widened joints) = limestone pavement.

5. Stalactite (roof, 'holds tight') vs stalagmite (ground); meet = pillar.

✓ Karst LACKS integrated surface drainage — water disappears underground.

✗ A uvala is smaller than a single doline.

✓ A uvala is formed by TWO OR MORE coalesced dolines (larger).

■ **MAINS ANGLE**

Explain how solution creates karst erosional and depositional landforms, and relate them to India's limestone cave systems and their fragile aquifers.

■ **STUDY LINK**

Geography → Physical → Karst/Limestone (GC Leong: Underground Water; Majid: Karst Geomorphology)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — Limestone Mining vs Karst Aquifers (Meghalaya)

GS-I / GS-III · Karst applied → Groundwater & environment

■ **NEWS TRIGGER**

Unregulated limestone mining for cement in Meghalaya threatens its karst caves and aquifers — risking cave collapse, disrupted underground streams and groundwater loss.

■ **INTRO & ORIGIN**

A karst aquifer stores water in solution-widened joints and caverns; mining the host limestone directly attacks this hidden water system.

■ **STATIC THEORY**

• **Karst aquifers** hold groundwater in cave/joint networks — a major water source but highly vulnerable.

• Cement-grade limestone mining causes **cave collapse**, destroys cave ecosystems and lowers groundwater recharge.

• Regulatory brakes: NGT/Supreme Court restrictions on mining in ecologically sensitive karst areas.

■ **MUST-KNOW FACTS**

1. Karst aquifer = groundwater in limestone caves/joints (vulnerable).

2. Meghalaya hosts India's largest cave systems (Siju, Krem Liat Prah).

3. Threat: cement/limestone mining -> cave collapse + aquifer damage.

■■ **UPSC TRAPS**

✗ Karst aquifers are well protected from contamination.

✓ They are HIGHLY vulnerable — pollutants travel fast through open joints/caves.

■ **MAINS ANGLE**

Link karst hydrology to the mining-vs-conservation conflict over Meghalaya's caves and aquifers.

■ **STUDY LINK**

Geography → Karst → Aquifers & Mining (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI